Fiscal Year:	FY 2007	**Task Last Updated:**	FY 07/25/2007
PI Name:	Hughes-Fulford, Millie Ph.D.		
Project Title:	Role of the Interleukin-2 Receptor in Signal Transduction		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	milliehf1@gmail.com	**Fax:**	FY 415-750-6667
PI Organization Type:	NON-PROFIT	**Phone:**	415-750-6940
Organization Name:	Northern California Institute for Research and Education		
PI Address 1:	Lab of Cell Growth, Mail Code 151F, VAMC		
PI Address 2:	4150 Clement Street		
PI Web Page:			
City:	San Francisco	**State:**	CA
Zip Code:	94121	**Congressional District:**	8
Comments:	PI is deceased as of February 2021		
Project Type:	Flight	**Solicitation / Funding Source:**	96-HEDS-04
Start Date:	01/15/1999	**End Date:**	09/30/2008
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA ARC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:	Soyuz/ISS		
Flight Assignment:	Soyuz (changed 7/07 per PI) ISS (changed 7/07 per PI)		
Key Personnel Changes/Previous PI:	Augusto Cogoli retired and the ESA PI is now Isabelle Walthers. The flight experiment on Soyuz was executed in orbit by ESA Astronaut Thomas Reiter. The incubation time was at 1.5 hours instead of the planned 4 hours. As a result, we had to make standards for the new timepoint. The gene array results should be completed by the end of summer.		
COI Name (Institution):	Walther, Isabelle (Swiss Federal Institute of Technology)		
Grant/Contract No.:	NAG2-1286		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	This work is collaboration between Professor Augusto Cogoli and Professor Millie Hughes-Fulford to study the role of the interleukin-2 receptor in signal transduction of T cells. The nearly total loss of T cell activation in real microgravity was discovered in a Spacelab in 1983 and was confirmed later by several experiments in space. Inhibition of IL-2R expression is probably one of the causes of such loss of activation. It is planned to analyze the pathway of mitogenic signal transduction in T cells in true microgravity as well as at variable G levels between 0 and 1xg. The objectives of this research project are: 1. To investigate selected critical steps of T cell activation 2. To test the hypothesis that a failure in the expression of the IL-2R is causing the loss of activity in microgravity 3. To identify windows of sensitivity to gravity during the activation process 4. To establish thresholds of gravi-sensing T cells. The main experimental approach will consist of the activation of cultures of purified peripheral blood cells with T cell mitogens in real and simulated microgravity. The specific IL-2R mRNA will be quantitatively determined with the reverse Transcriptase-polymerase chain reaction (RT-PCR) technology, the insertion of IL-2R in the membrane will be visualized by immunofluorscence and its secretion in the supernatant will be measured by immunoassay. Signal transduction will be furthered analyzed by determining the early expression of oncogenes. The Laboratory of Cell Growth (LCG), headed by Dr. Millie Hughes-Fulford 's lab will analyse the gene expression in T cells and determine which genes require gravity for full induction..
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	All terrestrial life began in a gravity field. This study of alterations of early T-cell activation microgravity gives us the unique opportunity to examine the role of Earth's gravity in immune function. In the late 60's returning Apollo astronauts were found to have reduced immune function after spaceflight, taking approximately 7 days to recover normal function. Later studies by Dr. Cogoli demonstrated changes in the T-cells during exposure to microgravity. This program will examine the expression of 30,000 genes and 10,00 ESTs including IL-2, and the three IL-2 receptor subunits (alpha, beta and gamma) in microgravity with and without an artificial 1g gravity vector and compare those results with ground controls. Results will be verified using qRTPCR. These studies will show the regulation of the earliest signals that cause the T-cell to activate and the role of normal earth's gravity in that signaling.
Task Progress:	Due to the loss of our LEUKIN experiment on STS-107, we re-flew LEUKIN on the Soyuz taxi to the ISS in the Fall of 2006. The preparation and ground based execution of the flight experiment went flawlessly, except that the incubation was only 1.5 hours instead of four hours long. The primary objective required a four hour incubation and therefore was lost due to the reduced incubation time. The experiment will re-fly as PADIAC in October 2008. Many times a problem can also offer an opportunity; we are using these sample to examine the gene expression changes that occur during early activation under true microgravity conditions. To that end, we had to make new 1.5 hour standards. Since there was no extra money for new gene chip studies and had to wait for Padiac funding to continue the gene chip analysis of the 1.5hr LEUKIN samples. PADIAC funds will arrive in the near future and we will complete the studies for the 1.5 hour LEUKIN samples.
Bibliography Type:	Description: (Last Updated: 03/10/2018)
Articles in Peer-reviewed Journals	Sevigny MB, Li CF, Alas M, Hughes-Fulford M. "Glycosylation regulates turnover of cyclooxygenase-2." FEBS Lett. 2006 Dec 11;580(28-29):6533-6. PMID: 17113084 , Dec-2006
Articles in Peer-reviewed Journals	Hughes-Fulford M, Rodenacker K, Jutting U. "Reduction of anabolic signals and alteration of osteoblast nuclear morphology in microgravity." J Cell Biochem. 2006 Oct 1;99(2):435-49. PMID: 16619267 , Oct-2006
Articles in Peer-reviewed Journals	Li CF, Hughes-Fulford M. "Fibroblast growth factor-2 is an immediate-early gene induced by mechanical stress in osteogenic cells." J Bone Miner Res. 2006 Jun;21(6):946-55. PMID: 16753025 , Jun-2006
Articles in Peer-reviewed Journals	Hughes-Fulford M, Li CF, Boonyaratanakornkit J, Sayyah S. "Arachidonic acid activates phosphatidylinositol 3-kinase signaling and induces gene expression in prostate cancer." Cancer Res. 2006 Feb 1;66(3):1427-33. PMID: 16452198 , Feb-2006
Articles in Peer-reviewed Journals	Boonyaratanakornkit JB, Cogoli A, Li CF, Schopper T, Pippia P, Galleri G, Meloni MA, Hughes-Fulford M. "Key gravity-sensitive signaling pathways drive T cell activation." FASEB J. 2005 Dec;19(14):2020-2. Epub 2005 Oct 6. PMID: 16210397 , Dec-2005
Articles in Peer-reviewed Journals	Hughes-Fulford M, Tjandrawinata RR, Li CF, Sayyah S. "Arachidonic acid, an omega-6 fatty acid, induces cytoplasmic phospholipase A2 in prostate carcinoma cells." Carcinogenesis. 2005 Sep;26(9):1520-6. Epub 2005 May 5. PMID: 15878913 , Sep-2005
Articles in Peer-reviewed Journals	Hughes-Fulford M, Sugano E, Schopper T, Li CF, Boonyaratanakornkit JB, Cogoli A. "Early immune response and regulation of IL-2 receptor subunits." Cell Signal. 2005 Sep;17(9):1111-24. Epub 2005 Feb 25. PMID: 15993752 , Sep-2005